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ORIGINAL PAPER

Detecting audio forgery using deep learning techniques with attention mechanisms on ResNet++

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Abstract

The emergence of deepfake technology, along with other forms of audio forgeries, raises new concerns for digital security and forensic examination. This paper proposes an effective audio forgery detection framework that utilizes a modified ResNet++ deep learning model with attention mechanisms and advanced preprocessing techniques. The algorithm includes a robust detection method that incorporates multi-scale feature fusion and transformer-based attention to a Mel-spectrogram. The model successfully overcomes challenges posed by intricate forgery patterns and overlapping background noise, demonstrating enhanced performance in noisy conditions. In addition, the inclusion of systematic preprocessing steps increases estimation accuracy, which is critical for real-time processing. The proposed algorithm shows good results, boasting 99.92% training accuracy, 95.45% validation accuracy, 99.52% precision, and a false positive rate of only 1.58%.

Keywords Audio Forgery Detection · Deepfake forgery · Transformer · ResNet++

1 Introduction

In the domain of audio forensics, deepfake audio forgery stands out as one of the most sophisticated and deceptive forms of threat. Such forgeries exploit advanced AI techniques and create highly authentic synthesized speech that mirrors the voice features of a chosen speaker. The capability to generate new unconstrained speech segments or modify pre-recorded speech in a realistic manner makes deepfake audio technology a powerful tool for misinformation, impersonation, and evading authentication systems that rely on voice recognition. Deepfake audio generation might create audio in which the speaker has supposedly said sentences they never actually said. This increases the level of difficulty in detection. The field of audio spoofing and its detection has significantly advanced. Deep neural networks introduced end-to-end learning from raw audio or time-frequency representations. Squeeze-and-Excitation (SE) blocks [4] further enhance feature maps by emphasizing salient patterns in spectrograms.

In [2], a CNN-based model reached 85.99% accuracy but suffered from severe overfitting, a common issue with limited fake and real samples. Recent work in [9] proposed a DNN with Human Log-Likelihoods (HLLs), but the model produced an Equal Error Rate (EER) of 12.24%, reflecting the ongoing challenge of generalizing across synthetic speech models.

Motivation and contribution

The rapid evolution of multimedia technologies has amplified the risk of audio forgeries, posing threats to privacy, security, and trust in communication. This work addresses three key challenges:

1. Deepfake audio often bypasses existing systems by mimicking authentic speech. We counter this with a multi-step preprocessing pipeline to restore the signal closer to its original form.
2. Current methods lack robustness to diverse real-world conditions. We use the ASV-Spoof dataset, which spans 7 recording environments, 26 recording devices, and 25 playback devices, to ensure broad generalization.

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3. Existing models fail to generalize across different attack types (e.g., TTS, VC, replay). Our ResNet++ architecture with integrated attention mechanisms effectively detects a wide range of forgery scenarios.

2 Proposed methodology

The proposed methodology provides a detailed pipeline for audio processing and classification of audio forgeries, with a primary focus on detecting deepfakes. It incorporates all steps from the preprocessing of raw audio data to feature extraction, design of deep neural network for classification, to the implementation of training technique. In addition, it also ensures robustness to variation in conditions so that generalization and performance in classification are enhanced. This method addresses the requisites of high classification performance for audio files through the application of sophisticated techniques for feature extraction, neural network design, and audio classification. The framework incorporates spectrogram representations of the audio, attention with transformer mechanisms, and global–local dependency capturing.

2.1 Preprocessing

Preprocessing plays a vital role in improving the quality of input features and ensuring the data is suitable for analysis by deep–learning models. The proposed pipeline includes normalization, noise reduction, mel-spectrogram extraction, pre–emphasis filtering, and augmentation techniques.

Normalization and Noise Reduction: Audio signals are normalized to have zero mean and unit variance, ensuring consistency across the dataset. Gaussian noise is suppressed using spectral gating techniques, improving clarity and enhancing feature quality.

Mel–Spectrogram Extraction: Audio signals are transformed into mel-spectrograms using the Short-Time Fourier Transform (STFT) with optimized parameters (128 mel bands, window size of 2048, hop length of 512). The spectrograms are converted to the decibel (dB) scale for perceptually meaningful amplitude variations. Histogram equalization is applied to enhance contrast, making spectral features more distinguishable.

Pre-Emphasis Filtering: High–frequency components are emphasized by applying a pre–emphasis filter, which improves the signal–to–noise ratio and enhances feature discrimination.

Augmentation Techniques: SpecAugment is applied to mel–spectrograms during training, including time masking (mask width = 20) and frequency masking (mask width = 10) to simulate real-world distortions. Spectrograms are resized to 224×224 resolution and normalized using ImageNet statistics to ensure dimensional consistency and improve model convergence. This preprocessing pipeline ensures robust feature extraction and prepares the data for effective classification by the ResNet++ architecture.

2.2 Data augmentation techniques

To enhance model generalization and robustness, several augmentation strategies are applied directly to mel–spectrogram representations:

SpecAugment: This technique introduces controlled distortions to simulate real-world conditions:

- **Time Masking**: Randomly masks segments along the time axis, simulating short-duration dropouts and teaching the model to recognize patterns across discontinuous time segments.
- **Frequency Masking**: Randomly masks frequency bands, helping the model become invariant to frequency-specific distortions and focus on global spectral patterns.

Image–Based Augmentations: Spectrograms are resized to a standard 224×224 resolution for dimensional consistency and normalized using ImageNet statistics to improve model convergence and leverage transfer learning. These augmentation techniques significantly improve the model's ability to generalize across diverse acoustic conditions, ensuring robust performance against unseen forgery scenarios.

2.3 Proposed ResNet++ architecture

Our proposed architecture is a new hybrid method of audio spoofing detection with improved deep learning. The proposed model employs a ResNet50 variant as a backbone, with domain-specific attention mechanisms added to enhance feature extraction from mel-spectrograms. Input spectrograms ($3 \times 224 \times 224$) are fed through initial convolutional layers prior to going to four residual blocks with deepening channels ($256 \rightarrow 512 \rightarrow 1024 \rightarrow 2048$) and diminishing spatial dimensions. One unique feature of this model is the integration of a Convolutional Block Attention Module (CBAM)[8] following the last residual block, which implements both channel and spatial attention mechanisms. The channel attention part selectively provides greater importance to important feature channels and reduces the impact of unimportant ones by employing average and max pooling strategies. At the same time, the spatial attention part identifies important regions in the feature maps by holistically using features pooled by both average and max operations across the channel direction. To effectively extract features at diverse scales, the architecture makes use of parallel processing streams that consist of a Squeeze-and-Excitation (SE) block and a Transformer block, both acting on the same 2048-channel feature

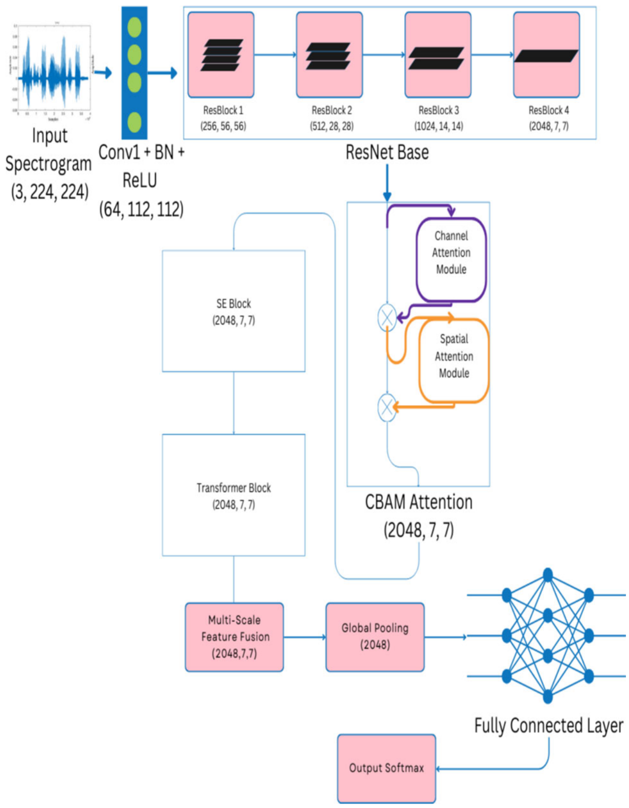

Fig. 1 ResNet++ Model Architecture Overview Diagram

maps. The SE block dynamically modulates channel-wise feature responses by learning inter-channel relationships. The Transformer block, on the other hand, utilizes self-attention mechanisms to learn long-range dependencies and thereby provides complementary global contextual information. The architecture is further supported by a Multi–Scale Feature Fusion module that combines features at varying receptive field sizes through dilated convolutions (rates 1, 2, and 4) in order to enhance the model's ability to recognize spoofing artifacts at various scales. Global pooling reduces the spatial dimensions to a small 2048-dimensional feature vector before end classification through fully connected layers with dropout regularization. The loss function combines cross-entropy with aspects of focal loss to tackle the class imbalance. The key components of the model architecture are shown in figure 1.

2.3.1 Input representation

The input to the network consists of normalized mel–spectrograms derived from raw audio signals. These spectrograms are processed using the Short-Time Fourier Transform (STFT) with a window size of 2048, a hop length of 512, and 128 mel bands. Each spectrogram is converted into a 3-channel image and resized to a resolution of 224×224, resulting in an input tensor of size (3, 224, 224)

Convolutional Backbone The backbone is based on ResNet-50, modified to enhance feature extraction. The initial convolutional layer has a kernel size of 7×7, stride 2, and padding 3, producing an output of size (64, 112, 112). This is followed by four residual stages, each consisting of bottleneck residual blocks. Stage 1 contains 3 blocks with an output size of (256, 56, 56). Stage 2 comprises 4 blocks, producing an output size of (512, 28, 28). Stage 3 consists of 6 blocks, resulting in an output size of (1024, 14, 14). Finally, Stage 4 includes 3 blocks with an output size of (2048, 7, 7). Residual connections are incorporated to facilitate gradient flow and mitigate the vanishing gradient problem, enabling the construction of a deeper network.

Attention Mechanisms To refine feature representations, the architecture integrates two attention mechanisms:

(i) Convolutional Block Attention Module (CBAM) [8] refines features through sequential channel and spatial attention. The channel attention mechanism employs a reduction ratio of 16, while the spatial attention uses a 7×7 convolutional kernel. Equation 1 represents the attention mechanism used to enhance feature representations by applying both channel and spatial attention. The operational process of channel attention map generation can be shown as below:

$$M_c = \sigma\Big(\mathrm{MLP}\big(\mathrm{AvgPool}(F_{\mathrm{base}})\big) + \mathrm{MLP}\big(\mathrm{MaxPool}(F_{\mathrm{base}})\big)\Big) \tag{1}$$

here, M_c denotes the attention map being generated and F_{base} represents the extracted features of the input image using ResNet-50. This computes attention weights across channels using global average and max pooling followed by a multi–layer perceptron (MLP) and a sigmoid activation function. After generating the channel attention map, we apply the channel attention which can be shown as below:

$$F_c = M_c \otimes F_{\mathrm{base}} \tag{2}$$

here, F_c represents the channel–refined feature map.The channel attention map is multiplied element-wise with the base feature map to enhance important channels. Now, we generate spatial attention map which captures spatial dependencies by applying a convolution over pooled feature maps.

$$M_s = \sigma(\mathrm{Conv}([\mathrm{AvgPool}(F_c); \mathrm{MaxPool}(F_c)])) \tag{3}$$

here, M_s represents the generated spatial attention map. **(ii) The spatial attention map** is applied to refine the attended feature map, enhancing crucial regions in the image.

$$F_{\mathrm{att}} = M_s \otimes F_c \tag{4}$$

Table 1 Validation accuracy for different Transformer hyperparameter configurations

Heads	2 ×	3×	4×	5×	6×
2	91.3	92.1	93.8	94.2	94.0
4	92.7	93.5	94.5	94.8	94.6
8	94.8	95.0	**95.7**	95.4	95.2
16	95.1	95.3	95.9	95.6	95.5

here, F_{att} represents the attention-refined feature map. Finally, squeeze-and-Excitation (SE) blocks are integrated into each residual block to adaptively scale channel-wise features, emphasizing informative regions.

2.3.2 Transformer blocks

To capture global dependencies and long-range interactions within the spectrograms, transformer blocks are applied to the feature map of size $(2048, 7, 7)$. The feature map is flattened to $(49, 2048)$, where 49 corresponds to the spatial dimensions, and processed through Multi–Head Self–Attention (MHSA) using 8 attention heads and Feed–Forward Network (FFN) hidden layer with size 8192 (4× the input size). Further the output is reshaped back to (2048, 7, 7) for subsequent processing.

The specific configuration of our transformer blocks is carefully determined through extensive hyperparameter tuning experiments aimed at balancing model capacity and generalization performance. The choice of 8 attention heads represents an optimal trade-off between computational efficiency and representational power. This configuration allows the model to simultaneously attend to different frequency patterns and temporal relationships in the audio spectrograms without excessive parameter overhead that could lead to overfitting.This has been depicted visually in the table 1.

Similarly, the feed-forward network's hidden dimension of 8192 (4× the input dimension) is selected based on empirical validation. Our experiments show that smaller hidden dimensions resulted in underfitting on complex forgery patterns, while larger dimensions increased the risk of overfitting without meaningful performance gains. The 4× ratio provided the best validation performance across multiple test scenarios. Additionally, we incorporated dropout layers (rate=0.1) within the transformer blocks to further regularize the model and prevent co-adaptation of features, which is particularly important given the high-dimensional representation space of the transformer.

2.3.3 Multi-scale feature fusion

To aggregate features across multiple scales, the architecture employs a multi-scale feature fusion module with parallel branches using convolutional kernels of sizes 1×1, 3×3, 5×5, and 7×7. The fused output has a size of $(2048, 7, 7)$, effectively capturing both local and global patterns.

2.3.4 Classification head

The fused features are passed through a global average pooling (GAP) layer, reducing the spatial dimensions from $(2048, 7, 7)$ to (2048). A fully connected layer with 512 units is applied, followed by batch normalization, ReLU activation, and dropout (0.5). Finally, a softmax layer maps the output to the desired number of classes, here as we have a binary classification of forged or genuine. The entire process is carried out through fully connected layers z = FC(v) where FC represents the fully connected layers. Finally, apply softmax to obtain probability distribution p = Softmax(z) Regarding the architectural details, the proposed ResNet++ model has an input size of (3, 224, 224) and consists of 50 convolutional layers (ResNet backbone) along with transformer, attention, and fusion layers. The total number of parameters in the model is approximately 23 million. So, the ResNet++ architecture integrates attention mechanisms, residual connections, transformer modules, and multi–scale feature fusion to effectively extract both local and global features from mel–spectrograms. These components work together synergistically, enabling robust and accurate audio classification. The following equation represents the classification decision based on the predicted probability distribution.

$$y = \arg\max_i (p_i) \tag{5}$$

where, p_i denotes the probability assigned to class i. arg max selects the class index with the highest probability and y is the predicted class label. If $y = 1$, the audio sample is classified as genuine; otherwise, it is spoofed. Based on the output probabilities, we make our decision regarding whether the audio file is genuine or spoofed.

3 Experimental results and discussion

Here, we elaborate on the results of our experiments. The implementations of the proposed model have been compared with other models like VGG, LGBM, CNN + LSTM, and executed on a laptop with a Quadro M1000M GPU.

In this work, we used the ASVspoof 2017 dataset [6], a benchmark for Automatic Speaker Verification (ASV) and anti-spoofing systems. It includes a wide range of authentic and spoofed audio samples, covering attack types such as Voice Conversion (VC) and speech synthesis.

Table 2 Evaluation metrics summary after 50 epochs of training

Evaluation Metrics	Value (in %)
Cross Entropy Training Loss	0.0196
Training Accuracy	99.92
Cross Entropy Validation Loss	0.0172
Validation Accuracy	95.45
Precision	99.52
F1 Score	95.12
False Positive Rate (FPR)	1.58
True Negative Rate (TNR)	98.42

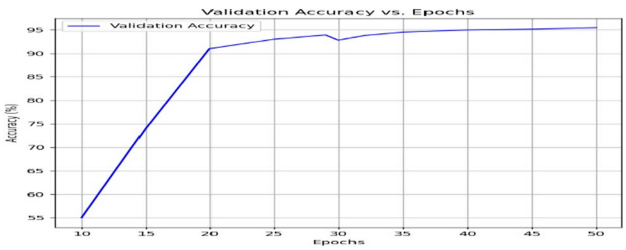

Fig. 2 Graphical representation of Accuracy vs Epochs

For this study, the training set contains 8000 audio samples and the validation set includes 3000. Each audio file is preprocessed to extract spectrogram features as model input. This setup ensures variability across genuine and spoofed samples. Interestingly, swapping the training and validation sets led to a 39% spike in validation accuracy, highlighting the dataset composition's strong influence on model generalization. Training on the larger set enabled the model to learn broader patterns, while validation on the smaller set restricted variability, making high accuracy easier to attain.

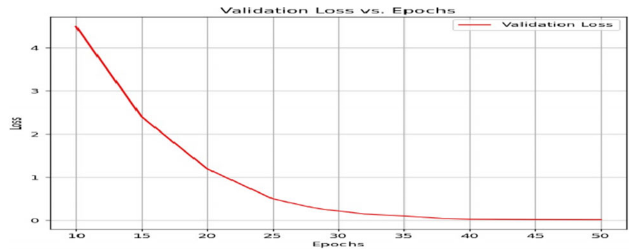

Fig. 3 Graphical representation of Cross Entropy Loss vs Epochs

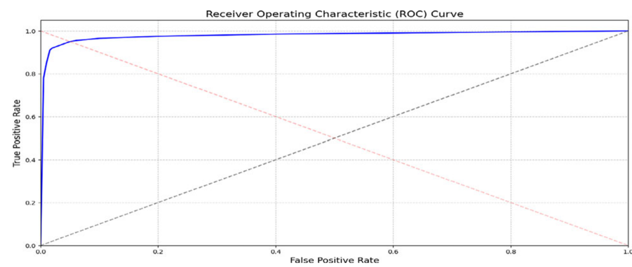

Fig. 4 ROC curve showing the trade-off between True Positive Rate and False Positive Rate, with AUC = 0.965

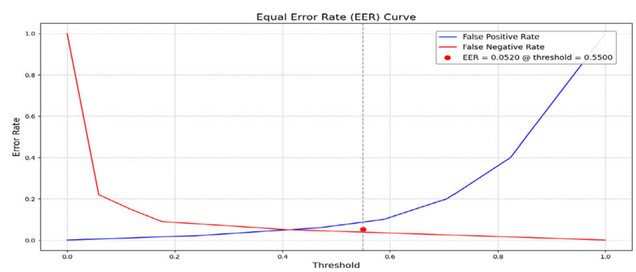

Fig. 5 Equal Error Rate (EER) curve showing the intersection of False Positive Rate and False Negative Rate curves, with EER = 5.2%

Performance evaluation

To evaluate the performance of the proposed model, several standard classification metrics have been employed, including accuracy, cross-entropy loss, precision, recall, F1 score, false positive rate (FPR), and true negative rate (TNR) (Table 2).The improvement in performance over epochs is illustrated in the accuracy vs. epoch (Fig. 2) and loss vs. epoch (Fig. 3) plots.

To provide deeper insights into the model's performance, we present advanced visualization metrics including ROC curves, confusion matrix, and Equal Error Rate (EER) analysis. Figure 4 shows the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. The Area Under the Curve (AUC) value of 0.965 indicates excellent discriminative capability, with the model achieving a TPR of 91.07% at an FPR of just 1.58%. Figure 5 presents the Equal Error Rate (EER) analysis, where the FPR and False Negative Rate (FNR) curves intersect. The EER value of 5.2% at a threshold of 0.55 represents the operating point where the false positive and false negative rates are equal. This relatively low EER further confirms the model's balanced performance in minimizing both types of errors.

To test the generalizability of our approach, we collected 400 genuine audio samples from students of SVNIT, representing natural variations in speaking styles, accents, and recording conditions. While the forged audio samples used for evaluation were sourced from the ASVspoof 2019 DF dataset which contains deepfake audio designed to simulate realistic spoofing scenarios. The confusion matrix in Figure 6 provides a detailed breakdown of the model's predictions. Out of 400 genuine samples, 392 were correctly classified (True Negatives), while only 8 were misclassified as fake (False Positives). Similarly, out of 400 fake samples, 375 were correctly identified (True Positives), with 25 misclassified as genuine (False Negatives). This demonstrates

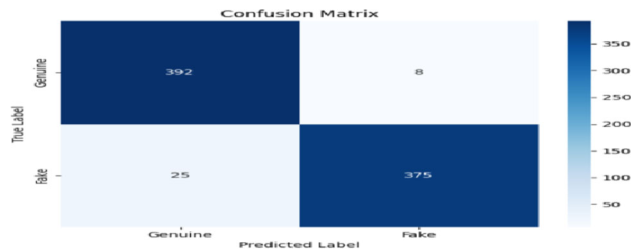

Fig. 6 Confusion matrix showing the distribution of predictions across genuine and fake audio samples in real world scenario

the model's strong performance in correctly identifying both genuine and fake audio samples.

Ablation studies

The individual contribution of each component in our ResNet++ architecture are as follows:

Attention Mechanisms Impact: Removing CBAM led to a 3.22% drop in validation accuracy and over 1% in precision, highlighting its role in focusing on key spectrogram regions. Eliminating SE blocks caused a 1.69% drop, confirming their importance in channel-wise feature recalibration. **Transformer Blocks Contribution**: The largest drop (4.33%) occurred without transformer blocks, emphasizing their role in modeling long-range dependencies and global context. **Multi-Scale Feature Fusion**: Removing this module reduced validation accuracy by 1.56%, showing its value in capturing features at multiple resolutions. **Data Augmentation Efficacy**: Disabling augmentation caused a 5.21% drop in accuracy and 5.48% in F1 score, underlining its importance in generalization. **Base Architecture Comparison**: The full ResNet++ outperformed ResNet50 with a 3.03% accuracy and 8.04% precision gain, validating the impact of the enhancements (Table 3).

Comparison with related work

We implemented our method alongside LSTM, ResNet50, VGG16, and CNN, comparing it with several well-established models. Our approach outperformed all others, achieving 99.92% training accuracy and 99.52% precision. The LSTM model performed poorly, with 59.48% training accuracy and 58.46% precision, highlighting the limitations of sequential models for this task. While CNN reached 99.20% training accuracy, our method still exceeded it in precision (99.52%) and F1 score (95.12%). These results confirm that our method is not only competitive but also more effective and reliable than the compared models (Table 4).

3.1 Edge deployment trade-offs

While ResNet++ delivers strong detection accuracy, its deployment on resource-constrained edge devices requires balancing efficiency and performance.

Challenges: The model's 23 million parameters create substantial computational overhead, increasing memory usage and inference latency, which may limit real-time applications.

Strategies for Optimization:

- **Model Compression**: Quantization and pruning can reduce memory footprint and computational load with minimal accuracy loss.
- **Hierarchical Processing**: A lightweight preliminary screening model (e.g., MobileNet) can be used before activating the full ResNet++ for suspicious samples.
- **Adaptive Deployment**: Dynamic complexity adjustment based on available resources can balance accuracy and efficiency.

Future work will focus on resource-aware variants to maintain acceptable performance thresholds while meeting edge deployment constraints.

4 Conclusion

This paper introduces a strong end-to-end audio forgery detection system utilizing new preprocessing, attention-based spectrogram features, and a light neural network. The model, even with a low-end consumer GPU, has a

Table 3 Ablation study results of the proposed architecture

Model Variant	Validation Accuracy (%)	Precision (%)	F1 Score (%)
Full ResNet++ (Ours)	94.45	99.52	94.20
Without CBAM	91.23	98.41	92.15
Without SE Blocks	92.76	98.95	93.05
Without Transformer Blocks	90.12	97.38	91.46
Without Multi-Scale Fusion	92.89	98.78	92.94
Without Data Augmentation	89.24	96.15	88.72
Base ResNet50	91.42	91.48	91.13

Table 4 Comparison with other state–of–art audio forgery detection techniques (in %)

Model	Train Accuracy	Validation Accuracy	Precision	F1 Score
Our Approach	99.92	95.45	99.52	95.12
SVM[5]	85.00	92.00	69.00	65.00
Random Forest[5]	62.00	80.00	60.00	57.00
KNN[5]	62.00	75.00	61.00	61.00
XGBoost[5]	59.00	70.00	59.00	57.00
LGBM[5]	60.00	75.00	60.00	58.00
ResNet50[10]	99.76	89.67	91.48	91.13
Asymmetric Bagging[1]	NA	88.00	67.00	81.00
VGG16[10]	99.55	93.36	NA	NA
CNN-LSTM[3]	NA	94.73	91.50	95.53

95.45% validation accuracy and 99.52% precision, reflecting minimal false positives. A dataset-swapping experiment illustrates the effects of dataset size and distribution on generalization. The architecture is found to be resistant to both pre- and post-processing attacks and is further enhanced through data augmentation and ensemble transformer integration. Experiments on FakeAVCeleb and audio from zero-shot generators Vall-E and Bark are planned as future work to test robustness to new generative techniques.

Author contributions Deep Das: Conducted the implementation and experiments; prepared the initial draft of the manuscript. Rahul Dixit: Supervised the work; provided guidance throughout the project and contributed to drafting and refining the manuscript.

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Data availability: No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known conflict of interest.

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